

ADVANCING CULTURAL ECONOMY AND LOCAL ECONOMIC EMPOWERMENT THROUGH AI-BASED VIRTUAL FITTING OF PAPUAN BATIK USING DETECTRON2 AND DENSEPOSE

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Abstract

Papuan Batik represents a valuable cultural heritage and an important economic asset that contributes to the development of Indonesia's cultural economy. Characterized by unique motifs inspired by indigenous philosophies, local biodiversity, and regional identities, Papuan Batik possesses significant potential to enhance the livelihoods of local artisans and Micro, Small, and Medium Enterprises (MSMEs). However, geographical isolation, limited market accessibility, inadequate digital promotion, and the absence of standardized cultural documentation continue to hinder its wider recognition and economic growth. These challenges restrict the participation of Papuan artisans in broader domestic and international markets, thereby limiting opportunities for sustainable local economic empowerment. This research proposes an AI-based virtual fitting platform utilizing Detectron2 and DensePose to support the digital transformation of Papuan Batik and strengthen its role within the cultural economy ecosystem. The platform provides a high-precision virtual fitting experience that enables users to visualize Papuan Batik apparel digitally before purchase, thereby increasing consumer engagement and expanding market reach. In addition, the system functions as a digital cultural repository by integrating historical narratives, philosophical meanings, and geographical origins of batik motifs, contributing to cultural preservation and intellectual property protection. To further promote local economic empowerment, the platform incorporates artisan information and location-based mapping features that facilitate direct connections between consumers and Papuan craftsmen, reducing intermediary barriers and creating broader economic opportunities. By integrating virtual fitting technology, cultural digitization, and artisan-market connectivity, this research demonstrates how artificial intelligence can be leveraged to advance cultural economy development and inclusive economic growth. The proposed platform aims to increase the economic value and global visibility of Papuan Batik, strengthen the competitiveness of local MSMEs, empower indigenous artisans through direct market participation, and establish a sustainable model for technology-driven cultural preservation and economic empowerment in remote regions.

Keyword: Cultural Economy, Inclusive Growth, Local Economic Empowerment, Papuan Batik, Virtual Fitting

Introduction

For the rest of the paper, please use 10-point Times New Roman. This document explains and demonstrates how to prepare your camera-ready manuscript for *the 9th International Seminar on Business, Economics, Social Science, and Technology (ISBEST) 2026 proceedings*. Please, read these instructions and follow outline of this text. The manuscripts must be written on A4 size paper with right-left-top-bottom margins of 3-3-3-3.

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Table 1. Cross Tab Analysis

Dimention	Generation	Poor	Normal	Good	Total
Technology Enable	Z	0.0%	17.1%	82.9%	100.0%
	Y	4.3%	7.1%	88.6%	100.0%
Interactivity	Z	0.0%	10.0%	90.0%	100.0%
	Y	2.9%	5.7%	91.4%	100.0%

Source: Data Processing Result, 2026

Special Signs

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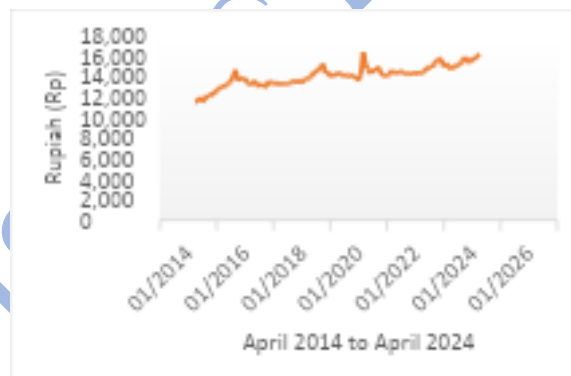


Figure 1. Graph of Indonesian Currency Exchange Rate

Equations

Equations (refer with: Eq. 1, Eq. 2, ...) should be indented 5 mm (0.2"). There should be one line of space above the equation and one line of space below it before the text continues. The equations have to be numbered sequentially, and the number put in parentheses at the right-hand edge of the text. Equations should be punctuated as if they were an ordinary part of the text. Punctuation appears after the equation but before the equation number, e.g.

$$c^2 = a^2 + b^2$$

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